Code : 100403

B.Tech 4th Semester Exam., 2022

(New Course)

DIGITAL ELECTRONICS

Time : 3 hours *Full Marks* : 70

Instructions :

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

*(iv) Question No. **1** is compulsory.*

1. Answer any *seven* of the following : 2×7=14

(a) What is the next number in the following octal counting sequence?

724, 725, 726, 727, _____

(b) What do you mean by a positive logic system and a negative logic system?

(c) Subtract using 9's complement :

745.81 – 436.62

(d) The following operation is correct for at least one number system. Find the correct number system :

1234 + 5432 = 6666

(e) What is a tri-state logic?

(f) Which are the fastest logic family and the slowest logic family?

(g) Which memory technology needs the least power?

(h) What is a register? What is a shift register?

(i) What is a master-slave flip-flop?

(j) Fill in the blank :

$(100101000111)_{(BCD)} = (_____)_{10}$

2. *(a)* Which of the following are analog quantities and which are digital?

Number of atoms in a sample of material, Altitude of an aircraft, Pressure in a bicycle tire, Current through a speaker, and Timer setting on a microwave oven.

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(b) What do you mean by self-complementing code? Write two self-complementing codes.

(c) If the waveforms *A* and *B* shown in Fig. 1 are applied to a two-input XOR gate, determine the output waveform :

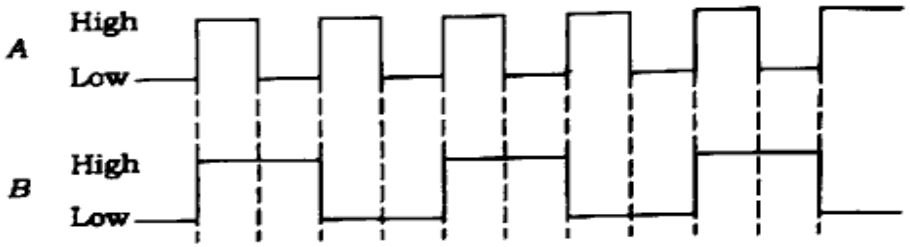


Fig. 1

4+4+6

3. (a) Perform the following in excess-3 code using the 10's complement method :

$$239 - 597$$

(b) Design and implement a 4-bit binary to Gray converter.

(c) Reduce the following expression using K-map and implement it in AOI logic as well as in NOR logic :

$$F = \Pi M(0, 1, 2, 3, 4, 7)$$

5+4+5

4. (a) Reduce the following expression and implement it using universal logic gate :

$$\Sigma m(1, 5, 6, 12, 13, 14) + d(2, 4)$$

(b) Use a multiplexer to implement the logic function $F = A \oplus B \oplus C$. 7+7

5. (a) Determine the *Q*-output waveform if the inputs shown in Fig. 2 are applied to the gated *D*-latch which is initially RESET :

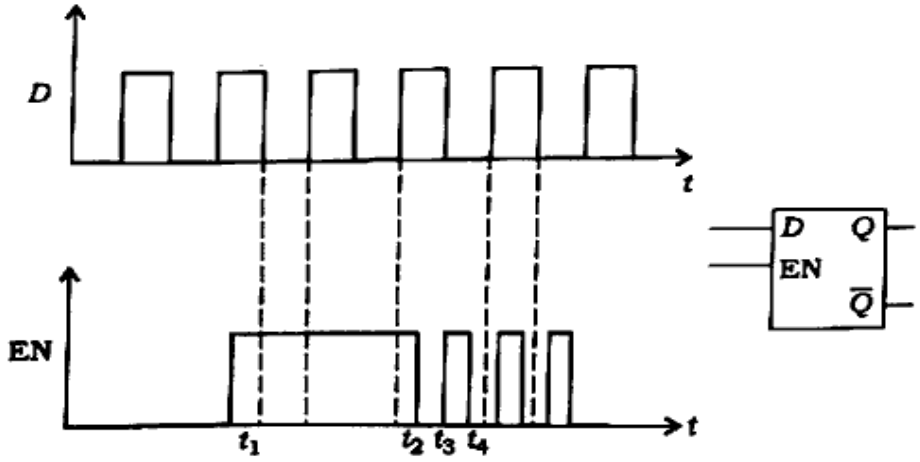


Fig. 2

(b) Design the conversion circuit for *S-R* flip-flop to *J-K* flip-flop. 7+7

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6. *(a)* What is basic difference between a counter and a shift register?

(b) With neat diagrams, explain the working of the following types of shift registers :

(i) Serial-in, serial-out

(ii) Serial-in, parallel-out

(c) Design and implement a mod-10 asynchronous counter using T flip-flops. 3+6+5

7. *(a)* Design and implement a synchronous 3-bit up/down counter using J-K flip-flops. https://www.akubihar.com

(b) Determine the resolution of—

(i) 6-bit DAC;

(ii) 12-bit DAC

in terms of percentage.

(c) What is the resolution of a 9-bit DAC which uses a ladder network? What is this resolution expressed as a percentage? If the full-scale output voltage of this converter is +5 V, what is resolution in volts? 5+4+5

8. *(a)* For the 4-bit weighted resistor DAC shown in Fig. 3, determine the—

(i) weight of each input bit if the inputs are 0 V and 5 V;

(ii) full-scale output if $R_f = R = 1\ \text{k}\Omega$.

Also, find the full-scale output if R_f is changed to 500 Ω :

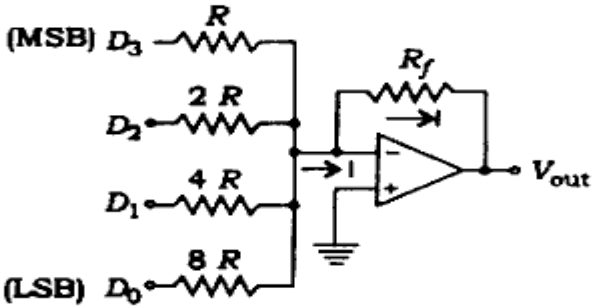


Fig. 3

(b) What is a PLD? What do '×' and 'dot' represent on a PLD diagram? 9+5

9. *(a)* What are the different technologies used for the fabrication of ROM memories? Determine how many 16K × 4 memory circuits would be required to construct each of the following memories :

(i) 256K × 8

(ii) 128K × 16

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(b) Define Memory cell, Address and Byte.

(c) Using the simplified connection format of a PLA, show how an 8 × 1 PROM should be programmed to implement the logic function

$F = \Sigma m(1, 4, 5, 7)$ 4½+4½+5

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